Total No. of Questions : 6]

SEAT No. :

**P86**

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# BE/Insem./APR-126
## B.E. (Mechanical)
## 402050 B : FINITE ELEMENT ANALYSIS
## (2012 Course) (Semester - II) (Elective - IV)

***Time : 1 Hour]*** ***[Max. Marks : 30***

***Instructions to the candidates:***

1) ***Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.***
2) ***Figures to the right side indicate full marks.***
3) ***Neat diagrams must be drawn wherever necessary.***
4) ***Use of scientific calculator allowed.***
5) ***Assume suitable data, if necessary.***

***Q1)*** a) Discuss : Basic steps of FEA. **[4]**

b) Derive the expression for shape function for a two noded bar element taking natural coordinate '$\xi$' where $(-1 \le \xi \le 1)$. **[6]**

OR

***Q2)*** a) What are advantages and Disadvantages of FEA. **[4]**

b) Write short note on Penalty Approach. **[6]**

***Q3)*** a) Explain the term 'Shape Functions'. Why polynomial terms are preferred for shape functions in finite element method? **[4]**

b) Derive an expression for the element stiffness matrix of the two noded truss element. **[6]**

OR

***Q4)*** a) Explain 'Displacement function'. **[4]**

b) Using direct stiffness method, determine the nodal displacement of stepped bar shown in Fig. 1 Assume $E_1 = 200GPa$, $A_1 = 150mm^2$, $L_1 = 50mm$, $E_2 = 70GPa$, $A_2 = 100\ mm^2$, $L_2 = 50mm$, $F_1 = 10KN$ & $F_2 = 5KN$. **[6]**

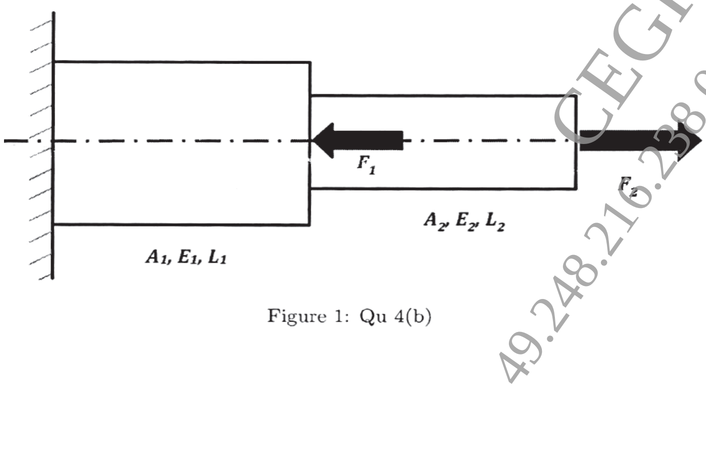


Figure 1: Qu 4(b)

***P.T.O.***

*Q5)* a) Explain the terms :

i) Constant strain triangle (CST). **[3]**

ii) Linear strain triangle (LST). **[3]**

b) Explain 'Pascals Triangle'. How it is used in decision of interpolation function in element formulations. **[4]**

OR

*Q6)* a) What do you mean by displacement function? Write down convergence criteria for finite element analysis. **[6]**

b) Explain the concept of Plane Stress and Plane Strain in Finite Element Method. **[4]**

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